ICS 253 – Discrete Structures
Chapter 2: Proofs

Question bank: 90 multiple choice + 10 written problems

Questions from ICS 253 old quizzes and homework (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

Which equation shows, directly from the definition of an odd integer, that -17 is odd?

(A) $-17 = 2(-8) - 1$

(B) $-17 = -(2 \cdot 8 + 1)$

(C) $-17 = 2(-9) + 1$

(D) $-17 = 2(-8.5)$

Answer: (C). The definition requires the form $2k + 1$ with k an integer; here $k = -9$.

My answer

Q2. [new]

Let k be an integer. Which expression is always an odd integer?

(A) $4k + 2$

(B) $k^2 + k$

(C) $2k + 4$

(D) $6k^2 + 3$

Answer: (D). $6k^2 + 3 = 2(3k^2 + 1) + 1$ and $3k^2 + 1$ is an integer. The other three are always even.

My answer

Q3. [new]

Which pair of integers has the same parity?

(A) -13 and 21

(B) 17 and -4

(C) 0 and 7

(D) $2k$ and $2k + 1$, where k is an integer

Answer: (A). $-13 = 2(-7) + 1$ and $21 = 2(10) + 1$ are both odd.

My answer

Q4. [new]

Which of the following is NOT a valid way to show that 0.375 is rational?

(A) $0.375 = \dfrac{3}{8}$

(B) $0.375 = \dfrac{375}{1000}$

(C) $0.375 = \dfrac{0.75}{2}$

(D) $0.375 = \dfrac{-6}{-16}$

Answer: (C). A rational number must be written as a/b with a, b integers and $b \neq 0$; 0.75 is not an integer.

My answer

Q5. [new]

Let $r = \frac{5}{7}$ and $s = -\frac{2}{3}$. Writing $r - s$ as a ratio of two integers gives

(A) $\frac{3}{4}$

(B) $\frac{1}{21}$

(C) $\frac{7}{21}$

(D) $\frac{29}{21}$

Answer: (D). $\frac{5}{7} + \frac{2}{3} = \frac{15+14}{21} = \frac{29}{21}$.

My answer

Q6. [new]

An integer n is odd if $n = 2k + 1$ for some integer k. Which value of k shows, by the definition, that -7 is odd?

(A) $k = -4$

(B) $k = -3$

(C) $k = 3$

(D) No integer k works, so -7 is not odd.

Answer: (A). $2(-4) + 1 = -7$. With $k = -3$ we get -5, and with $k = 3$ we get 7.

My answer

Q7. **[new]**

Using the same definition of divides, which statement is FALSE?

(A) $-3 \mid 12$

(B) $0 \mid 0$

(C) $7 \mid 7$

(D) $1 \mid n$ for every integer n

Answer: (B). The definition requires the divisor to be nonzero, so $0 \nmid 0$.

My answer

Q8. **[new]**

A real number r is rational if $r = \frac{a}{b}$ for some integers a and b with $b \neq 0$. Which choice shows, by the definition, that 0.25 is rational?

(A) $a = 1$, $b = 4$

(B) $a = 25$, $b = 0$

(C) $a = 0.25$, $b = 1$

(D) $a = 4$, $b = 1$

Answer: (A). $\frac{1}{4} = 0.25$ with integers $a = 1$ and $b = 4 \neq 0$. (B) divides by 0, (C) uses a non-integer, and (D) gives 4.

My answer

Q9. **[new]**

Which statement is TRUE?

(A) 1 is neither prime nor composite.

(B) 1 is prime.

(C) 1 is composite.

(D) 91 is prime.

Answer: (A). Primes and composites must be greater than 1. Also $91 = 7 \cdot 13$ is composite.

My answer

Q10. [new]

Which of the following numbers is composite?

(A) 211

(B) 223

(C) 227

(D) 221

Answer: (D). $221 = 13 \cdot 17$, so 13 is a divisor with $1 < 13 < 221$. The others are prime.

My answer

Q11. [new]

The statement "It is not true that $x \leq -2$" is equivalent to

(A) $x < -2$

(B) $x \geq -2$

(C) $x \geq 2$

(D) $x > -2$

Answer: (D). If $x \leq -2$ is false, then neither $x < -2$ nor $x = -2$ holds, so $x > -2$.

My answer

Q12. [new]

"The real number x is not positive" is equivalent to

(A) $x < 0$

(B) $x \geq 0$

(C) $x \leq 0$

(D) $x \neq 0$

Answer: (C). Positive means $x > 0$; its negation is $x \leq 0$ (nonpositive).

My answer

Q13. [new]

Which conditional statement is true for every real number x?

(A) If $x \geq 3$, then $x > 3$.

(B) If $x > 3$, then $x \geq 3$.

(C) If x is nonnegative, then x is positive.

(D) If $x^2 > 4$, then $x > 2$.

Answer: (B). $x > 3$ implies "$x > 3$ or $x = 3$". (A) and (C) fail at $x = 3$ and $x = 0$; (D) fails at $x = -3$.

My answer

Q14. [new]

Which of the following is an existential statement?

(A) The product of two negative real numbers is a positive real number.

(B) If a real number x satisfies $x^2 < 1$, then $-1 < x < 1$.

(C) Every integer is either even or odd.

(D) There is a perfect square that is the sum of two nonzero perfect squares.

Answer: (D). It asserts that some object with a property exists. The others are universal statements.

My answer

Q15. HW3 Rosen 1.8-1

To prove "$n^2 + 1 \geq 2n$ for every positive integer n with $1 \leq n \leq 4$" by exhaustion, which facts must be checked?

(A) Only $n = 1$ and $n = 4$, the end points.

(B) $2 \geq 2,\ 5 \geq 4,\ 10 \geq 6,\ 17 \geq 8$

(C) The inequality for $n = 0, 1, 2, 3, 4$.

(D) None; the statement involves squares, so exhaustion cannot be used.

Answer: (B). The domain is $\{1, 2, 3, 4\}$; check $n = 1, 2, 3, 4$: $2 \geq 2$, $5 \geq 4$, $10 \geq 6$, $17 \geq 8$.

My answer

Q16. HW2 Almarhoon (Rosen 1.7-20)

Let $P(n)$ be the proposition "If n is a positive integer, then $n^2 \geq n$." Which argument correctly proves $P(1)$?

(A) 1 is a positive integer and $1^2 = 1 \geq 1$, so the conclusion holds and $P(1)$ is true.

(B) $P(1)$ is true because its hypothesis is false.

(C) $P(1)$ is false because 1^2 is not greater than 1.

(D) $P(1)$ can only be proved by checking every positive integer.

Answer: (A). The conclusion $1^2 \geq 1$ is verified directly (a trivial proof); the hypothesis is true, so (B) is wrong.

My answer

Q17. HW3 Rosen 1.7-21

Let $P(n)$ be "If a and b are positive real numbers, then $(a+b)^n \geq a^n + b^n$." Which statement about $P(1)$ is correct?

(A) $P(1)$ is false, since $a+b$ is not strictly greater than $a+b$.

(B) $P(1)$ holds only when $a = b$.

(C) $P(1)$ is true, since $(a+b)^1 = a+b = a^1 + b^1$, so the inequality holds with equality.

(D) $P(1)$ is true only if $a > b$.

Answer: (C). For $n = 1$ both sides equal $a+b$, and "$\geq$" allows equality.

My answer

Q18. [new]

Which is a valid beginning of a proof by universal generalization of "The difference of two odd integers is even"?

(A) Let $x = 3$ and $y = 5$.

(B) Suppose $x - y$ is even.

(C) Let x be an odd integer and let $y = x$.

(D) Let x and y be odd integers.

Answer: (D). Name arbitrary objects with only the assumptions in the theorem; (A) is an example, (C) adds an assumption, (B) assumes the conclusion.

My answer

Q19. Quiz 2 Alrasasi

Prove or disprove: For any real numbers x and y, if $x < y$, then $x^2 < y^2$.

(A) True for all real numbers.

(B) False; counterexample $x = 3$, $y = -1$.

(C) False; counterexample $x = 1$, $y = 3$.

(D) False; counterexample $x = -3$, $y = 1$.

Answer: (D). $-3 < 1$ but $9 \not< 1$. In (B) the hypothesis fails; in (C) the conclusion holds.

My answer

Q20. Quiz 2 Adel

Determine whether the statement is true or false: "If x and y are positive real numbers and $x < y$, then $x^2 < y^2$."

(A) False; $x = -3$, $y = 1$ is a counterexample.

(B) False; $x = y = 1$ is a counterexample.

(C) True only when x and y are integers.

(D) True; multiplying $x < y$ by $x > 0$ gives $x^2 < xy$, and by $y > 0$ gives $xy < y^2$, so $x^2 < y^2$.

Answer: (D). Direct proof as stated. $x = -3$ is not positive and $x = y$ violates $x < y$, so neither is a counterexample.

My answer

Q21. [new]

Which value of n is a counterexample to "For every integer $n > 1$, $2^n > n^2$"?

(A) $n = 1$

(B) $n = 3$

(C) $n = 5$

(D) $n = 6$

Answer: (B). $2^3 = 8 < 9 = 3^2$. $n = 1$ fails the hypothesis; $32 > 25$ and $64 > 36$.

My answer

Q22. [new]

Which value is a counterexample to "For every real number x, if $x > 0$ and $x < 4$, then $x^2 < 9$"?

(A) $x = 3$

(B) $x = 5$

(C) $x = -4$

(D) $x = 1$

Answer: (A). $0 < 3 < 4$ (both hypotheses hold) and $3^2 = 9 \not< 9$. 5 and -4 fail a hypothesis; 1 satisfies the conclusion.

My answer

Q23. Quiz 2 Alrasasi

Which choice proves that there are two real numbers x and y such that $\sqrt{x+y} = \sqrt{x} + \sqrt{y}$?

(A) $x = 1, y = 1$

(B) $x = 4, y = 9$

(C) $x = 0, y = 1$

(D) $x = 1, y = 3$

Answer: (C). $\sqrt{0+1} = 1 = \sqrt{0} + \sqrt{1}$. The others give $\sqrt{2} \neq 2$, $\sqrt{13} \neq 5$, $2 \neq 1 + \sqrt{3}$.

My answer

Q24. [new]

Which integer gives a constructive proof that "some integer can be written as the sum of the squares of two positive integers in two different ways"?

(A) 25

(B) 13

(C) 65

(D) 41

Answer: (C). $65 = 1^2 + 8^2 = 4^2 + 7^2$. $25 = 3^2 + 4^2$, $13 = 2^2 + 3^2$, $41 = 4^2 + 5^2$ in one way only.

My answer

Q25. HW3 Rosen 1.8-17

Suppose a and b are odd integers with $a \neq b$. Which integer c satisfies $|a - c| = |b - c|$?

(A) $c = a + b$

(B) $c = \frac{a+b}{2}$

(C) $c = \frac{a-b}{2}$

(D) $c = ab$

Answer: (B). c must be the midpoint. Since $a + b$ is even (sum of two odds), $c = (a + b)/2$ is an integer, and it is the only solution because $a \neq b$.

My answer

Q26. HW3 Rosen 1.8-19

If n is an odd integer, the integer k such that n is the sum of $k - 2$ and $k + 3$ is

(A) $k = \frac{n-1}{2}$

(B) $k = \frac{n+1}{2}$

(C) $k = n - 1$

(D) $k = 2n + 1$

Answer: (A). $(k - 2) + (k + 3) = 2k + 1 = n$ gives $k = (n - 1)/2$, an integer because n is odd.

My answer

Q27.

HW3 Rosen 1.8-32

To show $x^2 + y^2 = z^2$ has infinitely many positive integer solutions, take $x = m^2 - n^2$, $y = 2mn$, $z = m^2 + n^2$. With $m = 3$, $n = 2$ the solution is

(A) $(3, 4, 5)$

(B) $(8, 15, 17)$

(C) $(7, 24, 25)$

(D) $(5, 12, 13)$

Answer: (D). $9 - 4 = 5$, $2 \cdot 3 \cdot 2 = 12$, $9 + 4 = 13$, and $25 + 144 = 169$. In general $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$, and each $m > n \geq 1$ gives a different solution.

My answer

Q28.

[new]

Which of the following is a valid disproof of "For every integer $n \geq 0$, $n^2 + n + 41$ is prime"?

(A) $n = 40$: $40^2 + 40 + 41 = 1681 = 41^2$, which is not prime.

(B) $n = 1$: $1 + 1 + 41 = 43$, which is prime.

(C) $n = 2$: $4 + 2 + 41 = 47$, which is prime.

(D) It cannot be disproved, because it holds for $n = 0, 1, \ldots, 10$.

Answer: (A). One counterexample disproves a universal statement, and $1681 = 41 \cdot 41$ is composite. Checking cases where it holds proves nothing.

My answer

Q29.

[new]

To show that "There is an integer n such that $n^2 + n$ is odd" is false, one must prove

(A) For every integer n, $n^2 + n$ is even.

(B) There is an integer n such that $n^2 + n$ is even.

(C) For every integer n, $n^2 + n$ is odd.

(D) $n^2 + n$ is even for $n = 1$.

Answer: (A). By De Morgan's law, $\neg \exists n\, P(n) \equiv \forall n\, \neg P(n)$, and every integer is even or odd.

My answer

Q30.

[new]

Which statement about methods of proof is TRUE?

(A) Showing a statement holds for a few cases proves a universal statement.

(B) One example where the statement fails disproves an existential statement.

(C) Providing one counterexample is sufficient to disprove a universal statement.

(D) Proof by exhaustion can be used for any infinite domain.

Answer: (C). One counterexample falsifies "for all". Existential statements are disproved by a universal argument.

My answer

Q31. [new]

Which of the following is NOT one of the allowed assumptions in the proofs of this chapter (it must be proven first)?

(A) The sum of two integers is an integer.

(B) If n is an integer, there is no integer strictly between n and $n+1$.

(C) The square of any real number is greater than or equal to 0.

(D) The product of two rational numbers is rational.

Answer: (D). Closure of the rationals under multiplication is a theorem to be proven; the others are listed as allowed assumptions.

My answer

Q32. [new]

Which proof step is valid?

(A) If n is an integer and $n > 3$, then $n \geq 4$.

(B) If x and y are real and $x > y$, then $x^2 > y^2$.

(C) If x is a positive real number, then $\frac{1}{x} > x$.

(D) If x, y, z are real and $xz < yz$, then $x < y$.

Answer: (A). There is no integer between 3 and 4. (B) fails for $x = 1, y = -2$; (C) fails for $x = 2$; (D) needs $z > 0$.

My answer

Q33. [new]

Fill in the blank with the best phrase: "The integer n is even. _____, $n = 2k$ for some integer k."

(A) By definition

(B) We will show that

(C) Suppose

(D) In other words, we must prove

Answer: (A). The equation follows from the definition of an even integer.

My answer

Q34. [new]

Which proof fragment contains an INVALID use of existential instantiation?

(A) Since x and y are both even, $x = 2k$ and $y = 2k$ for some integer k.

(B) Since x is even, $x = 2k$ for some integer k. Since y is even, $y = 2j$ for some integer j.

(C) Since $x \mid y$, $x \neq 0$ and $y = mx$ for some integer m.

(D) Since r is rational, $r = a/b$ for some integers a, b with $b \neq 0$.

Answer: (A). Reusing the name k forces $x = y$, which is not given. Each instantiation needs a new name.

My answer

Q35. [new]

"If n is an odd integer, then $n = 2k + 1$ for some integer k. Let $n^2 = 2m + 1$ for some integer m. Since n^2 is two times an integer plus 1, n^2 is odd." The error in this argument is

(A) generalizing from examples.

(B) circular reasoning.

(C) assuming a fact proven elsewhere without reference.

(D) an invalid use of the word "Let".

Answer: (B). It assumes $n^2 = 2m + 1$, which is exactly the fact to be proven.

My answer

Q36. [new]

"4 is even and $4^2 = 16$ is even. 6 is even and $6^2 = 36$ is even. Therefore, if n is even then n^2 is even." The error is

(A) circular reasoning.

(B) skipping steps.

(C) using the same variable name twice.

(D) generalizing from examples.

Answer: (D). Checking a few elements does not prove a universal statement over an infinite domain.

My answer

Q37. [new]

"Suppose r is a rational number. The product of any two rational numbers is rational. Therefore r^2 is rational." What is wrong with this proof?

(A) It assumes a fact that has not been proven.

(B) It is circular reasoning.

(C) It generalizes from examples.

(D) Nothing; the proof is complete.

Answer: (A). Closure of rationals under multiplication has not been established in the proof.

My answer

Q38. [new]

Theorem: The product of an even integer and an odd integer is even.
Proof. "Since x is even, $x = 2k$ for some integer k. Since y is odd, $y = 2j + 1$ for some integer j. Then $xy = 2k(2j + 1) = 2(2kj + k)$, and $2kj + k$ is an integer, so xy is even. ■"
Which best practice is violated?

(A) The proof does not end with an end-of-proof symbol.

(B) The algebra is incorrect.

(C) The variables x and y are used without being introduced.

(D) The proof uses existential instantiation incorrectly.

Answer: (C). It should start with "Let x be an even integer and y an odd integer."

My answer

Q39. [new]

In a direct proof that "if x and y are integers and $x \mid y$, then $x \mid y^2$", we have $y = kx$ with $x \neq 0$. To finish we write $y^2 = mx$, where the integer m is

(A) k^2

(B) k^2x

(C) kx^2

(D) x^2

Answer: (B). $y^2 = k^2x^2 = (k^2x)x$, and k^2x is an integer.

My answer

Q40. HW3 Rosen 1.7-35

Consider these steps for solving $\sqrt{x+3} = 3 - x$: (1) given; (2) $x + 3 = x^2 - 6x + 9$ by squaring; (3) $0 = x^2 - 7x + 6$; (4) $0 = (x - 1)(x - 6)$; (5) $x = 1$ or $x = 6$. Which statement is correct?

(A) All steps are reversible, so both 1 and 6 are solutions.

(B) Step (4) is a factoring error.

(C) Step (5) is wrong because $ab = 0$ does not imply $a = 0$ or $b = 0$.

(D) Each step follows from the previous one, but squaring is not reversible; $x = 6$ is extraneous ($\sqrt{9} = 3 \neq -3$), so only $x = 1$ is a solution.

Answer: (D). The steps show only that a solution must be 1 or 6; checking, $x = 1$ works and $x = 6$ does not.

My answer

Q41. [new]

Which are the correct assumptions in a direct proof of "The difference between two distinct real numbers is not equal to zero"?

(A) Let x and y be real numbers.

(B) Let x and y be real numbers such that $x \neq y$.

(C) Let x and y be real numbers such that $x - y = 0$.

(D) Let x and y be real numbers such that $x > 0$ and $y > 0$.

Answer: (B). The hypothesis "distinct" means $x \neq y$; we then prove $x - y \neq 0$.

My answer

Q42. HW3 Rosen 1.7-4

In a direct proof that the additive inverse (negative) of an even number is even, we let $n = 2k$ for an integer k. The best concluding step is

(A) $-n = 2k - 2n$, so $-n$ is even.

(B) $-n = -2k + 0$, so $-n$ is even.

(C) $-n = 2(-k)$, and $-k$ is an integer, so $-n$ is even.

(D) $-n = 2k + 1$, so $-n$ is even.

Answer: (C). This writes $-n$ as 2 times an integer, matching the definition of even.

My answer

Q43. [new]

In a direct proof that the product of two odd integers is odd, with $x = 2k + 1$ and $y = 2j + 1$, we write $xy = 2m + 1$ where m equals

(A) $2kj + k + j$

(B) $4kj + 2k + 2j$

(C) $kj + k + j$

(D) $2kj + 1$

Answer: (A). $(2k+1)(2j+1) = 4kj + 2k + 2j + 1 = 2(2kj + k + j) + 1$.

My answer

Q44.

HW3 Rosen 1.7-7

A direct proof that every odd integer is the difference of two squares writes $n = 2k + 1$ as

(A) $k^2 - (k-1)^2$

(B) $(k+1)^2 - (k-1)^2$

(C) $(2k)^2 - (2k-1)^2$

(D) $(k+1)^2 - k^2$

Answer: (D). $(k+1)^2 - k^2 = 2k + 1 = n$. The other choices give $2k - 1$, $4k$, $4k - 1$.

My answer

Q45.

[new]

In the direct proof that the sum of two rational numbers $r = a/b$ and $s = c/d$ is rational, $r + s = \frac{ad+bc}{bd}$. Which fact justifies that the denominator is valid?

(A) $ad + bc$ is an integer.

(B) b and d are nonzero integers, so bd is a nonzero integer.

(C) a and c are nonzero.

(D) r and s are positive.

Answer: (B). A rational number needs an integer denominator that is not 0; $bd \neq 0$ because $b \neq 0$ and $d \neq 0$.

My answer

Q46.

[new]

A direct proof that $\frac{x}{y} + \frac{y}{x} \geq 2$ for all positive reals x, y starts from which true fact?

(A) $(x+y)^2 \geq 0$

(B) $x + y > 0$

(C) $(x-y)^2 \geq 0$

(D) $xy > 1$

Answer: (C). $(x-y)^2 \geq 0 \Rightarrow x^2 + y^2 \geq 2xy$; dividing by $xy > 0$ gives $\frac{x}{y} + \frac{y}{x} \geq 2$.

My answer

Q47. [new]

Theorem: The average of the squares of two even integers is an even integer. With $x = 2k$, $y = 2j$, the average equals

(A) $2(k^2 + j^2)$

(B) $4(k^2 + j^2)$

(C) $k^2 + j^2$

(D) $2(k + j)^2$

Answer: (A). $\frac{4k^2+4j^2}{2} = 2(k^2 + j^2)$, two times an integer.

My answer

Q48. [new]

"If a, b, c are integers and $a \mid bc$, then $a \mid b$ or $a \mid c$." This statement is

(A) true, by a direct proof.

(B) false; $a = 2$, $b = 4$, $c = 1$ is a counterexample.

(C) false; $a = 3$, $b = c = 3$ is a counterexample.

(D) false; $a = 4$, $b = c = 2$ is a counterexample.

Answer: (D). $4 \mid 4 = 2 \cdot 2$ but $4 \nmid 2$. In (B) and (C), a divides b, so they are not counterexamples.

My answer

Q49. [new]

In a direct proof that "if $a \mid b$ and $b \mid c$, then $a \mid c$", we have $b = ka$ and $c = jb$. Then

(A) $c = (j + k)a$

(B) $c = (j/k)a$

(C) $a = (jk)c$

(D) $c = (jk)a$, and jk is an integer, so $a \mid c$.

Answer: (D). Substituting $b = ka$ into $c = jb$ gives $c = j(ka) = (jk)a$, and $a \neq 0$.

My answer

Q50. [new]

Which of the following statements is FALSE?

(A) The average of two odd integers is an integer.

(B) The sum of two odd integers is even.

(C) The average of two odd integers is odd.

(D) The product of two odd integers is odd.

Answer: (C). Counterexample: 1 and 3 have average 2. (A) is true: $\frac{(2k+1)+(2j+1)}{2} = k + j + 1$.

My answer

Q51. [new]

Theorem: If x is a real number and $x \geq 3$, then $x^2 - 4x + 3 \geq 0$. Which is a correct direct argument?

(A) $x^2 - 4x + 3 = (x - 1)(x - 3)$; since $x - 1 \geq 2 > 0$ and $x - 3 \geq 0$, the product is ≥ 0.

(B) Since $x \geq 3$, $x^2 \geq 9 \geq 4x$, so the expression is positive.

(C) Check $x = 3$: $9 - 12 + 3 = 0 \geq 0$. Hence it holds for all $x \geq 3$.

(D) Assume $x^2 - 4x + 3 < 0$; then $x < 3$.

Answer: (A). (B) is false ($x^2 \geq 4x$ fails at $x = 3$), (C) generalizes from an example, (D) is not a direct proof.

My answer

Q52. HW2 Almarhoon (Rosen 1.7-26)

In proving "if n is even, then $7n + 4$ is even" (one half of "n is even iff $7n + 4$ is even"), we let $n = 2k$. Then $7n + 4 = 2m$ with $m =$

(A) $14k + 4$

(B) $7k + 4$

(C) $7k + 2$

(D) $14k + 2$

Answer: (C). $7(2k) + 4 = 14k + 4 = 2(7k + 2)$.

My answer

Q53. HW3 Rosen 1.7-32

To show "$3x - 1$ rational $\Rightarrow$ x rational", suppose $3x - 1 = \frac{p}{q}$ with p, q integers and $q \neq 0$. Then x equals

(A) $\frac{p-q}{3q}$

(B) $\frac{p+q}{3q}$

(C) $\frac{p+1}{3q}$

(D) $\frac{3p}{q} + 1$

Answer: (B). $3x = \frac{p}{q} + 1 = \frac{p+q}{q}$, so $x = \frac{p+q}{3q}$ with $3q \neq 0$.

My answer

Q54.

Quiz 1 Sec3 Ver A (232)

A proof by contrapositive of "If n^2 is divisible by 3, then n is divisible by 3" proves which statement?

(A) If n^2 is not divisible by 3, then n is not divisible by 3.

(B) If n is divisible by 3, then n^2 is divisible by 3.

(C) If n is not divisible by 3, then n^2 is not divisible by 3.

(D) If n is divisible by 3, then n^2 is not divisible by 3.

Answer: (C). The contrapositive of $p \to q$ is $\neg q \to \neg p$. (A) is the inverse, (B) the converse.

My answer

Q55.

[new]

In a proof by contrapositive of "For every integer n, if $3n + 7$ is odd, then n is even", we assume $n = 2k + 1$. The key computation is

(A) $3n + 7 = 6k + 10 = 2(3k + 5)$, so $3n + 7$ is even.

(B) $3n + 7 = 6k + 8 = 2(3k + 4)$, so $3n + 7$ is even.

(C) $3n + 7 = 6k + 10$, so $3n + 7$ is odd.

(D) $3n + 7 = 2(3k) + 7$, so n is even.

Answer: (A). $3(2k + 1) + 7 = 6k + 10 = 2(3k + 5)$, which proves the negation of the hypothesis.

My answer

Q56.

[new]

Consider: "For every real number x, if $x^5 + 7x^3 + 5x \geq x^4 + x^2 + 8$, then $x \geq 0$." A proof by contrapositive starts by assuming

(A) $x \geq 0$

(B) $x^5 + 7x^3 + 5x < x^4 + x^2 + 8$

(C) $x^5 + 7x^3 + 5x \geq x^4 + x^2 + 8$ and $x < 0$

(D) $x < 0$

Answer: (D). Assume the negation of the conclusion ($x < 0$) and prove $x^5 + 7x^3 + 5x < x^4 + x^2 + 8$. (C) is the start of a proof by contradiction.

My answer

Q57. [new]

What is proven in a proof by contrapositive of "If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$"?

(A) If $xy < 0$, then $x < 0$ and $y < 0$.

(B) If $xy < 0$, then $x < 0$ or $y < 0$.

(C) If $x < 0$ or $y < 0$, then $xy < 0$.

(D) If $xy \geq 0$, then $x \geq 0$ and $y \geq 0$.

Answer: (B). By De Morgan, $\neg(x \geq 0 \wedge y \geq 0) \equiv (x < 0) \vee (y < 0)$.

My answer

Q58. Quiz 2 Alrasasi

Let x and y be real numbers. To prove "if $x + y > 20$, then $x > 10$ or $y > 10$" by contrapositive, we

(A) assume $x > 10$ or $y > 10$ and show $x + y > 20$.

(B) assume $x \leq 10$ or $y \leq 10$ and show $x + y \leq 20$.

(C) assume $x + y \leq 20$ and show $x \leq 10$ and $y \leq 10$.

(D) assume $x \leq 10$ and $y \leq 10$; adding gives $x + y \leq 20$.

Answer: (D). $\neg(x > 10 \vee y > 10) \equiv x \leq 10 \wedge y \leq 10$, and adding the inequalities gives $x + y \leq 20$.

My answer

Q59. HW3 Rosen 1.7-15

Use a proof by contraposition to show that if $x + y \geq 2$, where x and y are real numbers, then $x \geq 1$ or $y \geq 1$. The correct proof is:

(A) Assume $x < 1$ and $y < 1$. Adding gives $x + y < 2$.

(B) Assume $x + y < 2$. Then $x < 1$ and $y < 1$.

(C) Assume $x \geq 1$ or $y \geq 1$. Then $x + y \geq 2$.

(D) Assume $x < 1$ or $y < 1$. Then $x + y < 2$.

Answer: (A). The negation of "$x \geq 1$ or $y \geq 1$" is "$x < 1$ and $y < 1$"; the conclusion $x + y < 2$ is the negated hypothesis.

My answer

Q60. Quiz 2 Adel

Prove by contrapositive: For every real number x, if x is irrational, then $-x$ is irrational. Which is the correct argument?

(A) Assume x is rational, $x = a/b$; then $-x = -a/b$ is rational.

(B) Assume $-x$ is rational, $-x = a/b$ with $b \neq 0$; then $x = (-a)/b$ is rational.

(C) Assume $-x$ is irrational; then x is irrational.

(D) Assume x is irrational and $-x$ is rational; this is a contradiction.

Answer: (B). Assume the negated conclusion ($-x$ rational) and derive the negated hypothesis (x rational); $-a$ and b are integers, $b \neq 0$.

My answer

Q61.

HW3 Rosen 1.7-13

Proving "if x is irrational, then $1/x$ is irrational" by contrapositive, we assume $1/x = a/b$ with a, b integers, $b \neq 0$, and conclude $x = b/a$. Why is $a \neq 0$?

(A) Because $b \neq 0$.

(B) Because x is irrational.

(C) Because $1/x \neq 0$ for every nonzero real x, so $a/b \neq 0$.

(D) Because every integer is nonzero.

Answer: (C). If $a = 0$ then $1/x = 0$, impossible. So $x = b/a$ is a ratio of integers with nonzero denominator, i.e. rational.

My answer

Q62.

[new]

Proof by contrapositive: "For every positive real x, if x is irrational, then $\sqrt{x}$ is irrational." Assuming $\sqrt{x} = a/b$ (a, b integers, $b \neq 0$), the next step is

(A) $x = a^2/b^2$, a ratio of integers with $b^2 \neq 0$, so x is rational.

(B) $x = a/b^2$, so x is rational.

(C) $x = \sqrt{a}/\sqrt{b}$, so x is rational.

(D) $\sqrt{x}$ is irrational, a contradiction.

Answer: (A). Squaring gives $x = a^2/b^2$; a^2, b^2 are integers and $b^2 \neq 0$.

My answer

Q63.

[new]

Theorem: If x is rational and y is irrational, then $x + y$ is irrational. Which is a valid start of a proof by contrapositive?

(A) Assume $x + y$ is irrational and x is rational; show y is irrational.

(B) Assume $x + y$ is rational; show x and y are both rational.

(C) Assume $x + y$ is rational and x is rational; show y is rational.

(D) Assume x is irrational and y is rational; show $x + y$ is rational.

Answer: (C). With two hypotheses, assume one hypothesis and the negated conclusion, and prove the other hypothesis is false.

My answer

Q64.

HW3 Rosen 1.7-27

For a positive integer n, to prove "if $5n + 6$ is odd, then n is odd" by contrapositive, we let $n = 2k$ and obtain

(A) $5n + 6 = 10k + 6 = 2(5k + 3) + 1$, odd.

(B) $5n + 6 = 10k + 6 = 2(5k + 3)$, even.

(C) $5n + 6 = 10k + 7 = 2(5k + 3) + 1$, odd.

(D) $5n + 6 = 5k + 6$, even.

Answer: (B). $5(2k) + 6 = 2(5k + 3)$, which is even. (The other direction: $n = 2k + 1$ gives $5n + 6 = 2(5k + 5) + 1$, odd.)

My answer

Q65.

HW3 Rosen 1.7-17

Show that if n is an integer and $n^3 + 5$ is odd, then n is even. In the proof by contraposition we let $n = 2k + 1$; then $n^3 + 5$ equals

(A) $2(4k^3 + 6k^2 + 3k + 2) + 1$

(B) $2(4k^3 + 6k^2 + 3k) + 1$

(C) $2(4k^3 + 6k^2 + 3k + 3)$

(D) $8k^3 + 5$

Answer: (C). $(2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 6$, which is even. (A proof by contradiction assumes $n^3 + 5$ odd and n odd and reaches the same computation.)

My answer

Q66.

[new]

Which statement is equivalent to the negation of "Among any pair of consecutive integers, at least one of the integers is even"?

(A) Among any pair of consecutive integers, both are odd.

(B) There is a pair of consecutive integers that are both odd.

(C) There is a pair of consecutive integers with exactly one even integer.

(D) Among any pair of consecutive integers, at least one is odd.

Answer: (B). $\neg\forall \equiv \exists\neg$, and "not at least one even" means "both odd".

My answer

Q67. [new]

A proof by contradiction of "The negative of any irrational number is irrational" begins by assuming

(A) every irrational number has a rational negative.

(B) there is a rational number whose negative is irrational.

(C) there is an irrational number whose negative is irrational.

(D) there is an irrational number whose negative is rational.

Answer: (D). The negation of "for every irrational x, $-x$ is irrational" is "some irrational x has $-x$ rational".

My answer

Q68. [new]

The theorem is $p \to c$. Which logical format describes a proof by contradiction?

(A) Assume p. Follow a series of steps to conclude c.

(B) Assume $\neg c$. Follow a series of steps to conclude $\neg p$.

(C) Assume $p \wedge \neg c$. Follow a series of steps to conclude $r \wedge \neg r$ for some proposition r.

(D) Assume c. Follow a series of steps to conclude p.

Answer: (C). (A) is a direct proof, (B) a proof by contrapositive, (D) proves the converse.

My answer

Q69. [new]

Which is a valid start to a proof by contradiction of "For every integer n, if n^2 is odd, then n is odd"?

(A) Suppose there is an integer n such that n^2 is odd and n is even.

(B) Suppose that for every integer n, n^2 is even.

(C) Suppose n is odd; we show n^2 is odd.

(D) Suppose there is an integer n such that n^2 is even and n is odd.

Answer: (A). $\neg\forall n\,(P(n) \to Q(n)) \equiv \exists n\,(P(n) \wedge \neg Q(n))$.

My answer

Q70. [new]

In the classic proof that $\sqrt{2}$ is irrational, we assume $\sqrt{2} = a/b$ in lowest terms and derive that a is even and then that b is even. What is the contradiction?

(A) $a^2 = 2b^2$ is impossible for integers.

(B) a and b have the common factor 2, contradicting that a/b has no common factors.

(C) b must be 0.

(D) $\sqrt{2}$ is greater than 1.

Answer: (B). We assumed no integer greater than 1 divides both a and b, but 2 divides both.

My answer

Q71. Quiz 1 Alenzi

Prove or disprove: The difference of two irrational numbers is irrational.

(A) True, by a proof by contradiction.

(B) True, since a rational minus an irrational is irrational.

(C) False; $\sqrt{2} - \sqrt{3}$ is a counterexample.

(D) False; $\sqrt{2} - \sqrt{2} = 0$ is a counterexample.

Answer: (D). $\sqrt{2}$ is irrational but $\sqrt{2} - \sqrt{2} = 0 = 0/1$ is rational. $\sqrt{2} - \sqrt{3}$ is irrational, so it is not a counterexample.

My answer

Q72. HW3 Rosen 1.7-11

Prove or disprove that the product of two irrational numbers is irrational.

(A) False: $\sqrt{2} \cdot \sqrt{2} = 2$, which is rational.

(B) True, since irrational numbers have no ratio form.

(C) False: $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$.

(D) True, by contrapositive.

Answer: (A). One counterexample disproves the universal statement; $\sqrt{6}$ is irrational, so (C) is not a counterexample.

My answer

Q73. Quiz 2 Adel

Prove that there is no smallest integer. A proof by contradiction proceeds as:

(A) Let $n = 0$; then $-1 < 0$, so there is no smallest integer.

(B) Assume n is the smallest integer. Then $n-1$ is an integer and $n-1 < n$, contradicting the choice of n.

(C) Assume n is the smallest integer. Then $n+1 > n$, a contradiction.

(D) Assume every integer n has $n-1 < n$; hence no smallest integer exists.

Answer: (B). (C) gives no contradiction; (A) is only an example; (D) does not assume the negation.

My answer

Q74.

HW3 Rosen 1.7-24

Show that at least three of any 25 days chosen must fall in the same month of the year. In a proof by contradiction, the key step is

(A) Each month has at least 28 days, so 25 days fit in one month.

(B) If each month had at least 3 chosen days, there would be 36 days.

(C) If at most 2 chosen days fell in each month, at most $2 \cdot 12 = 24$ days would be chosen, contradicting 25.

(D) $25/12 > 2$, so some month has exactly 3 days.

Answer: (C). Negation: every month has at most 2 of the chosen days, giving at most $24 < 25$ days.

My answer

Q75.

[new]

Theorem: There are infinitely many prime numbers. A proof by contradiction begins by assuming

(A) there are no prime numbers.

(B) every integer is prime.

(C) there are infinitely many composite numbers.

(D) there are only finitely many primes, $p_1, p_2, \ldots, p_n$.

Answer: (D). The negation of "infinitely many" is "finitely many".

My answer

Q76.

HW3 Rosen 1.7-39

Prove that at least one of the real numbers $a_1, a_2, \ldots, a_n$ is greater than or equal to their average A. Which proof is correct?

(A) Contradiction: if $a_i < A$ for all i, then $a_1 + \cdots + a_n < nA = a_1 + \cdots + a_n$, which is impossible.

(B) Direct: $a_1 \geq A$ always.

(C) Exhaustion: check $n = 1, 2, 3$.

(D) Contradiction: if $a_i > A$ for all i, then the sum exceeds nA.

Answer: (A). Negation: every $a_i < A$. Adding the n inequalities gives $nA < nA$.

My answer

Q77. [new]

Theorem: Every triangle has at least one acute angle. A proof by contradiction assumes a triangle with angles a, b, c has no acute angle. The contradiction is

(A) $a + b + c < 180$

(B) $a + b + c \geq 270$, but the angles of a triangle sum to 180

(C) $a = b = c = 60$

(D) one angle must be obtuse

Answer: (B). No acute angle means each angle is at least 90, so $a + b + c \geq 270 > 180$.

My answer

Q78. [new]

To adapt the proof that $\sqrt{2}$ is irrational to show $\sqrt{3}$ is irrational, which fact replaces "if n^2 is even, then n is even"?

(A) If n^2 is odd, then n is odd.

(B) If n is divisible by 3, then n^2 is divisible by 3.

(C) If n^2 is divisible by 3, then n is divisible by 3.

(D) If n^2 is divisible by 9, then n is divisible by 9.

Answer: (C). From $a^2 = 3b^2$ we need $3 \mid a$, then $b^2 = 3k^2$ gives $3 \mid b$, a common factor.

My answer

Q79. [new]

Theorem: There is no smallest positive real number. Suppose r is the smallest positive real number. Which number gives the contradiction?

(A) $r - 1$

(B) r^2

(C) $2r$

(D) $r/2$

Answer: (D). $r/2$ is positive and $r/2 < r$. ($r - 1$ need not be positive; $r^2 < r$ only when $r < 1$.)

My answer

Q80. [new]

A proof by contrapositive of $p \to c$ can be recast as a proof by contradiction. Which description is correct?

(A) Assume $p \wedge \neg c$; use $\neg c$ to derive $\neg p$; reach the contradiction $p \wedge \neg p$.

(B) Assume $\neg p$; derive $\neg c$; reach $c \wedge \neg c$.

(C) Assume c; derive p; reach $p \wedge \neg p$.

(D) This is impossible: contrapositive and contradiction are unrelated methods.

Answer: (A). A proof by contrapositive is a special case of proof by contradiction.

My answer

Q81. [new]

Which division of the domain is NOT a valid set of cases for a proof?

(A) x is an integer. Case 1: $x \leq 0$. Case 2: $x \geq 1$.

(B) x is a real number. Case 1: $x < 0$. Case 2: $x > 0$.

(C) x is a real number. Case 1: x is rational. Case 2: x is irrational.

(D) x is a real number. Case 1: $x \leq 0$. Case 2: $x \geq 0$.

Answer: (B). The value $x = 0$ is not covered. (A) is valid since no integer lies between 0 and 1; overlap as in (D) is allowed.

My answer

Q82. [new]

In a proof by cases that $x^2 - x$ is even for every integer x, the case "x is odd" ($x = 2k + 1$) gives

(A) $x^2 - x = 4k^2 + 2k = 2(2k^2 + k)$

(B) $x^2 - x = 4k^2 + 4k = 2(2k^2 + 2k)$

(C) $x^2 - x = 4k^2 + 2k + 1$

(D) $x^2 - x = 2k^2 + k$

Answer: (A). $(2k+1)^2 - (2k+1) = 4k^2 + 4k + 1 - 2k - 1 = 4k^2 + 2k$.

My answer

Q83. [new]

Theorem: Every perfect square n^2 is a multiple of 4 or a multiple of 4 plus 1. In the case $n = 2k + 1$, the best expression is

(A) $n^2 = 2(2k^2 + 2k) + 1$

(B) $n^2 = 4k^2 + 4k + 1$

(C) $n^2 = 4(k^2 + k) + 1$

(D) $n^2 = 4(k^2 + k + 1)$

Answer: (C). It shows n^2 as 4 times the integer $k^2 + k$, plus 1. (B) is equal but not in the required form.

My answer

Q84. [new]

In a proof by cases that $|xy| = |x|\,|y|$, consider the case $x \geq 0$ and $y < 0$. Which argument is correct?

(A) $xy \geq 0$, so $|xy| = xy = |x|\,|y|$.

(B) $xy \leq 0$, so $|xy| = -xy = x(-y) = |x|\,|y|$.

(C) $xy < 0$, so $|xy| = xy = x(-y)$.

(D) $|x| = -x$ and $|y| = y$, so $|x|\,|y| = -xy$.

Answer: (B). With $x \geq 0$, $y < 0$: $|x| = x$, $|y| = -y$, $xy \leq 0$, and $|xy| = -xy$ (also correct when $xy = 0$).

My answer

Q85. [new]

Proof: "Without loss of generality, assume x is even. Then $x = 2k$ and $xy = 2(ky)$, so xy is even." This proves "if x is even or y is even, then xy is even". Why is "without loss of generality" justified?

(A) Because the case "y is even" is identical with the roles of x and y swapped.

(B) Because x is always even.

(C) Because the case "both odd" never occurs in the integers.

(D) Because one case is enough to prove any theorem.

Answer: (A). The two cases are symmetric, so proving one suffices.

My answer

Q86. HW3 Rosen 1.8-5

To prove $\min(x, y) = \frac{x+y-|x-y|}{2}$ for all real x, y, we assume without loss of generality that $x \leq y$. Then

(A) $|x - y| = x - y$, so the expression equals y.

(B) $|x - y| = x + y$, so the expression equals 0.

(C) $|x - y| = y - x$, so the expression equals $\frac{x+y-(y-x)}{2} = x = \min(x, y)$.

(D) $|x - y| = y - x$, so the expression equals y.

Answer: (C). Since $x - y \leq 0$, $|x - y| = y - x$. The case $y \leq x$ is the same with x, y swapped (both sides are symmetric). Similarly $\max(x, y) = \frac{x+y+|x-y|}{2}$.

My answer

Q87.

HW3 Rosen 1.8-4

Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ for real a, b, c. Which set of cases suffices?

(A) Case 1: a is the smallest of the three; Case 2: b is the smallest; Case 3: c is the smallest.

(B) Case 1: $a \leq b$; Case 2: $b \leq c$.

(C) Case 1: a, b, c all positive; Case 2: all negative.

(D) One case: $a < b < c$.

Answer: (A). Some value is the smallest; in each case both sides equal it (e.g. if $a \leq b$ and $a \leq c$, both sides are a). (B) and (C) do not cover all orderings, (D) is a single example.

My answer

Q88.

[new]

Proving "for every integer n, n^2+3n+5 is odd" by cases: in the case $n = 2k+1$, we get $n^2+3n+5 = 2m+1$ with $m =$

(A) $2k^2 + 3k + 2$

(B) $2k^2 + 5k + 4$

(C) $4k^2 + 10k + 9$

(D) $2k^2 + 5k + 5$

Answer: (B). $(2k + 1)^2 + 3(2k + 1) + 5 = 4k^2 + 10k + 9 = 2(2k^2 + 5k + 4) + 1$. (For $n = 2k$: $2(2k^2 + 3k + 2) + 1$.)

My answer

Q89.

[new]

In a proof by cases that $|x| \geq x$ for every real number x, which argument handles the case $x < 0$?

(A) $|x| = x$, so $|x| \geq x$.

(B) $|x| = -x$ and $-x < 0 < x$.

(C) $|x| = -x$; since $x < 0$, $-x > 0 > x$, so $|x| > x$.

(D) $x^2 \geq 0$, so $|x| \geq x$.

Answer: (C). For $x < 0$, $|x| = -x > 0 > x$. (In the case $x \geq 0$, $|x| = x$.)

My answer

Q90. **[new]**

Proof by cases that $x^2 \geq 0$ for every real x. In the case $x < 0$, the correct justification is

(A) multiplying both sides of $x < 0$ by x keeps the direction, so $x^2 < 0$.

(B) multiplying both sides of $x < 0$ by the negative number x reverses the inequality, so $x \cdot x > 0 \cdot x$, i.e. $x^2 > 0$.

(C) $x^2 = |x|$, which is positive.

(D) $x^2 = -x \cdot x$, which is positive.

Answer: (B). Multiplying an inequality by a negative number reverses it. Together with the cases $x > 0$ and $x = 0$ this gives $x^2 \geq 0$.

My answer

Part B: Written problems

Q91. Quiz 2 Alkhulaifi

Prove the following statement: If x, y and z are integers such that $x \mid (y+z)$ and $x \mid y$, then $x \mid z$.

Answer:

Proof (direct). Let x, y, z be integers such that $x \mid (y+z)$ and $x \mid y$. We will show that $x \mid z$.

Since $x \mid (y+z)$, by definition $x \neq 0$ and there is an integer k such that $y+z = kx$. Since $x \mid y$, there is an integer m such that $y = mx$ (a new name, since k and m need not be equal).

Subtracting the second equation from the first gives

$$z = (y+z) - y = kx - mx = (k-m)x.$$

Since k and m are integers, $k-m$ is an integer. So z equals x times an integer and $x \neq 0$. Therefore $x \mid z$. ■

My answer

Q92. HW3 Rosen 1.7-5

Prove that if $m+n$ and $n+p$ are even integers, where m, n and p are integers, then $m+p$ is even. What kind of proof did you use?

Answer:

Proof (direct). Let m, n, p be integers such that $m+n$ and $n+p$ are even. We will show that $m+p$ is even.

By the definition of even, $m+n = 2k$ for some integer k, and $n+p = 2j$ for some integer j. Adding the two equations:

$$\begin{aligned}(m+n)+(n+p) &= 2k+2j \\ m+p+2n &= 2k+2j \\ m+p &= 2k+2j-2n = 2(k+j-n).\end{aligned}$$

Since k, j, n are integers, $k+j-n$ is an integer. So $m+p$ is 2 times an integer, and $m+p$ is even. ■

This is a *direct proof*: we assumed the hypothesis and derived the conclusion.

My answer

Q93. Quiz 2 Sec3 Ver A (232); HW2 Almarhoon (Rosen 1.7-18)

Prove that if n is an integer and $3n + 2$ is even, then n is even, using (a) a proof by contrapositive, and (b) a proof by contradiction.

Answer:

(a) Proof by contrapositive. Let n be an integer. We assume that n is odd (the negation of the conclusion) and prove that $3n + 2$ is odd (the negation of the hypothesis).

Since n is odd, $n = 2k + 1$ for some integer k. Then

$$3n + 2 = 3(2k + 1) + 2 = 6k + 5 = 2(3k + 2) + 1.$$

Since k is an integer, $3k + 2$ is an integer, so $3n + 2$ is odd. This proves the contrapositive, and so the original statement. ■

(b) Proof by contradiction. Suppose, for contradiction, that there is an integer n such that $3n+2$ is even and n is not even, i.e. n is odd. Then $n = 2k+1$ for some integer k, and as in (a), $3n+2 = 2(3k+2)+1$ is odd.

Since $3n + 2$ is also even, $3n + 2 = 2j$ for some integer j. Then $2j = 2(3k + 2) + 1$, so $j - (3k + 2) = \frac{1}{2}$. The left side is an integer (integers are closed under subtraction), but $\frac{1}{2}$ is not an integer. This is a contradiction. So n must be even. ■

My answer

Q94. Quiz 2 Alkhulaifi

Prove the following statement by contrapositive: For every pair of real numbers x and y, if $x^3 + xy^2 \leq x^2y + y^3$, then $x \leq y$.

Answer:

Proof (contrapositive). Let x and y be real numbers. Assume $x > y$ (the negation of $x \leq y$). We will show that $x^3 + xy^2 > x^2y + y^3$.

First, factor the difference:

$$x^3 + xy^2 - x^2y - y^3 = x(x^2 + y^2) - y(x^2 + y^2) = (x - y)(x^2 + y^2).$$

Since $x > y$, we have $x - y > 0$.

Next, $x^2 \geq 0$ and $y^2 \geq 0$, so $x^2 + y^2 \geq 0$. If $x^2 + y^2 = 0$, then $x^2 = y^2 = 0$, so $x = y = 0$, which contradicts $x > y$. Hence $x^2 + y^2 > 0$.

Multiplying the inequality $x - y > 0$ by the positive number $x^2 + y^2$ gives $(x - y)(x^2 + y^2) > 0$, that is,

$$x^3 + xy^2 - x^2y - y^3 > 0 \quad \Longrightarrow \quad x^3 + xy^2 > x^2y + y^3.$$

This is the negation of the hypothesis, so the contrapositive holds and the statement is proven. ■

My answer

Q95. Quiz 2 Alkhulaifi

Prove the following statement: If x is a real number such that $x^2 + 2x - 3 < 0$, then $-3 < x < 1$.

Answer:

Proof (contrapositive, with cases). Let x be a real number. The conclusion "$-3 < x < 1$" means "$x > -3$ and $x < 1$". By De Morgan's law its negation is "$x \leq -3$ or $x \geq 1$". Assume $x \leq -3$ or $x \geq 1$. We will show that $x^2 + 2x - 3 \geq 0$. Note that

$$x^2 + 2x - 3 = (x + 3)(x - 1).$$

Case 1: $x \leq -3$. Adding 3 gives $x + 3 \leq 0$. Adding -1 to $x \leq -3$ gives $x - 1 \leq -4 < 0$. Multiplying $x + 3 \leq 0$ by the negative number $x - 1$ reverses the inequality: $(x + 3)(x - 1) \geq 0$.

Case 2: $x \geq 1$. Then $x - 1 \geq 0$, and adding 3 to $x \geq 1$ gives $x + 3 \geq 4 > 0$. The product of a nonnegative number and a positive number is nonnegative: $(x + 3)(x - 1) \geq 0$.

In both cases $x^2 + 2x - 3 \geq 0$, which is the negation of the hypothesis. Therefore the contrapositive is true, and so is the statement. ■

My answer

Q96. Quiz 2 Alkhulaifi

Prove the following statement by contradiction: The number $2 - \sqrt{2}$ is irrational. (You may use the fact that $\sqrt{2}$ is irrational.)

Answer:

Proof (contradiction). Suppose, for contradiction, that $2 - \sqrt{2}$ is rational. Then $2 - \sqrt{2} = \frac{a}{b}$ for some integers a and b with $b \neq 0$. Solving for $\sqrt{2}$:

$$\sqrt{2} = 2 - \frac{a}{b} = \frac{2b - a}{b}.$$

Since a and b are integers, $2b - a$ is an integer, and $b \neq 0$. So $\sqrt{2}$ is a ratio of two integers with a nonzero denominator, i.e. $\sqrt{2}$ is rational. This contradicts the fact that $\sqrt{2}$ is irrational. Therefore the assumption is false, and $2 - \sqrt{2}$ is irrational. ■

My answer

Q97.

Quiz 1 Alenzi

Prove that there are no integers n and m such that $n^2 = 4m + 2$.

Answer:

Lemma. If n is an integer and n^2 is even, then n is even.
Proof (contrapositive). If n is odd, $n = 2k + 1$ for an integer k, and $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd. □

Proof (contradiction). Suppose there are integers n and m with $n^2 = 4m+2$. Then $n^2 = 2(2m+1)$, and $2m + 1$ is an integer, so n^2 is even. By the lemma, n is even, so $n = 2k$ for some integer k. Substituting,

$$4k^2 = 4m + 2 \quad \Longrightarrow \quad 2k^2 = 2m + 1 \quad \Longrightarrow \quad k^2 - m = \tfrac{1}{2}.$$

Since k and m are integers, $k^2 - m$ is an integer, but $\frac{1}{2}$ is not an integer. (Equivalently: $2k^2$ is even and $2m + 1$ is odd, and they cannot be equal.) This is a contradiction. Hence no such integers n and m exist. ■

My answer

Q98.

Quiz 2 Alrasasi

Use a proof by contradiction to prove: "If n is an integer and $n^3 + 3$ is odd, then n is even."

Answer:

Proof (contradiction). The statement is $\forall n\,(n^3 + 3 \text{ odd} \to n \text{ even})$. Its negation is

$$\exists n\,\big((n^3 + 3 \text{ is odd}) \wedge (n \text{ is odd})\big).$$

So suppose there is an integer n such that $n^3 + 3$ is odd and n is odd. Since n is odd, $n = 2k + 1$ for some integer k. Then

$$n^3 + 3 = (2k + 1)^3 + 3 = 8k^3 + 12k^2 + 6k + 1 + 3 = 2(4k^3 + 6k^2 + 3k + 2).$$

Let $a = 4k^3 + 6k^2 + 3k + 2$, an integer since k is an integer. So $n^3 + 3 = 2a$ is even.

But $n^3 + 3$ is also odd, so $n^3 + 3 = 2b + 1$ for some integer b. Then $2a = 2b + 1$, giving $a - b = \frac{1}{2}$, which is impossible since $a - b$ is an integer. This contradiction shows the assumption is false, so if $n^3 + 3$ is odd then n is even. ■

My answer

Q99. HW3 Rosen 1.7-12

Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

Answer:

The statement is **true**.

Proof (contradiction). Let r be a nonzero rational number and x an irrational number. Since r is rational, $r = \frac{a}{b}$ for integers a, b with $b \neq 0$; since $r \neq 0$, also $a \neq 0$.

Suppose, for contradiction, that rx is rational. Then $rx = \frac{c}{d}$ for integers c, d with $d \neq 0$. Since $r \neq 0$ we can divide by r:

$$x = \frac{c/d}{a/b} = \frac{cb}{da}.$$

Here cb and da are integers, and $da \neq 0$ because $d \neq 0$ and $a \neq 0$. So x is rational, which contradicts the assumption that x is irrational. Therefore rx is irrational. ■

(The condition "nonzero" is needed: $0 \cdot \sqrt{2} = 0$ is rational.)

My answer

Q100. HW3 Rosen 1.8-7

Prove the triangle inequality: if x and y are real numbers, then $|x| + |y| \geq |x + y|$, where $|t| = t$ if $t \geq 0$ and $|t| = -t$ if $t < 0$.

Answer:

Lemma. For every real number t: $t \leq |t|$ and $-t \leq |t|$.
Proof (cases). Case 1: $t \geq 0$. Then $|t| = t$, so $t \leq |t|$; and $-t \leq 0 \leq t = |t|$. Case 2: $t < 0$. Then $|t| = -t$, so $-t \leq |t|$; and $t < 0 < -t = |t|$. □

Proof (cases on the sign of $x + y$). Let x and y be real numbers.

Case 1: $x + y \geq 0$. Then $|x + y| = x + y$. By the lemma, $x \leq |x|$ and $y \leq |y|$; adding the two inequalities,

$$|x + y| = x + y \leq |x| + |y|.$$

Case 2: $x + y < 0$. Then $|x + y| = -(x + y) = (-x) + (-y)$. By the lemma, $-x \leq |x|$ and $-y \leq |y|$; adding,

$$|x + y| = (-x) + (-y) \leq |x| + |y|.$$

Every real $x + y$ is either ≥ 0 or < 0, so the two cases cover all possibilities. Therefore $|x| + |y| \geq |x + y|$ for all real x, y. ■

My answer